Linear Algebra

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Chapter 1

Matrix Multiplication

1.1 Matrix Multiplication

Matrix multiplication combines two matrices, A and B, to produce a new matrix, C. This process involves three main steps:

1.1.1 Compatibility Check

For multiplication, the number of columns in A (matrix 1) must match the number of rows in B (matrix 2). If A is $m \times n$ and B is $n \times p$, the result, C, will be $m \times p$.

1.1.2 Dot Product Calculation

Element C_{ij} in matrix C is calculated by taking the dot product of the ith row of A and the jth column of B:

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

This dot product is repeated for each element in C, combining rows of A with columns of B.

Consider matrices A and B where A is a 2×3 matrix and B is a 3×2 matrix.
Let

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

and

$$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

.
The resulting matrix C, from the dot product of A and B, will be a 2×2 matrix. To calculate the first element C_{11}, we take the dot product of the first row of A and the first column of B:

$$C_{11} = a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31}$$

Similarly, to calculate the second element in the first row of C, C_{12}, we take the dot product of the first row of A and the second column of B:

$$C_{12} = a_{11} \cdot b_{12} + a_{12} \cdot b_{22} + a_{13} \cdot b_{32}$$

Continuing this process for the second row of C, we find:

$$C_{21} = a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31}$$
$$C_{22} = a_{21} \cdot b_{12} + a_{22} \cdot b_{22} + a_{23} \cdot b_{32}$$

Thus, the complete matrix C can be expressed as:

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$$

This example shows how the dot product is utilized to multiply two matrices, with each element C_{ij} in the resultant matrix C being the sum of products of corresponding elements from the ith row of A and the jth column of B.

1.1.3 Example 1

Consider two matrices A and B where A is a 2×2 matrix and B is a 2×2 matrix.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

The product AB is calculated as:

$$AB = \begin{pmatrix} 1(2) + 2(1) & 1(0) + 2(3) \\ 3(2) + 4(1) & 3(0) + 4(3) \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 10 & 12 \end{pmatrix}$$

1.1.4 Example 2

Consider A as a 2×3 matrix and B as a 3×2 matrix.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

The product AB is:

$$AB = \begin{pmatrix} 1(1) + 2(2) + 3(3) & 1(4) + 2(5) + 3(6) \\ 4(1) + 5(2) + 6(3) & 4(4) + 5(5) + 6(6) \end{pmatrix} = \begin{pmatrix} 14 & 32 \\ 32 & 77 \end{pmatrix}$$

1.2 Hadamard Product

The Hadamard product is a simpler, element-wise multiplication of two matrices of the same size, A and B, resulting in a new matrix C:

$$C_{ij} = A_{ij} \cdot B_{ij}$$

This operation requires A and B to be of equal dimensions.

1.2.1 Example 1

For the Hadamard product, matrices A and B must be of the same dimensions. Let both be 2×2 matrices.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

The Hadamard product $A \odot B$ is:

$$A \odot B = \begin{pmatrix} 1(5) & 2(6) \\ 3(7) & 4(8) \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 21 & 32 \end{pmatrix}$$

1.2.2 Example 2

If A and B are 2×3 matrices:

$$A = \begin{pmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

The Hadamard product $A \odot B$ is:

$$A \odot B = \begin{pmatrix} -1(1) & 0(2) & 1(3) \\ 2(4) & -2(5) & 2(6) \end{pmatrix} = \begin{pmatrix} -1 & 0 & 3 \\ 8 & -10 & 12 \end{pmatrix}$$

Chapter 2

Understanding Linear Dependence and Independence

2.1 Introduction

Linear dependence and independence are foundational concepts in linear algebra, essential for understanding vectors, vector spaces, and the solutions to linear systems.

2.2 Definitions

Linear Independence: A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in a vector space is said to be linearly independent if no vector in the set can be written as a linear combination of the others. In mathematical terms, the vectors are linearly independent if the only solution to the equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{0}$ is $c_1 = c_2 = \ldots = c_n = 0$.
Linear Dependence: Conversely, a set of vectors is considered linearly dependent if at least one of the vectors can be expressed as a linear combination of the others. This means there exists a non-trivial combination of these vectors that equals the zero vector $\mathbf{0}$, where not all coefficients c_i are zero.

2.3 Examples

2.3.1 Linear Independence

Consider vectors $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. These vectors are linearly independent because there is no scalar k such that $k\mathbf{a} = \mathbf{b}$ or $k\mathbf{b} = \mathbf{a}$.

2.3.2 Linear Dependence

Now, let $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{w} = 2\mathbf{v} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$. Vectors $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent because $\mathbf{w}$ is a scalar multiple of $\mathbf{v}$.

2.4 Graphical Representation

To visualize these concepts, we refer to the figures below.

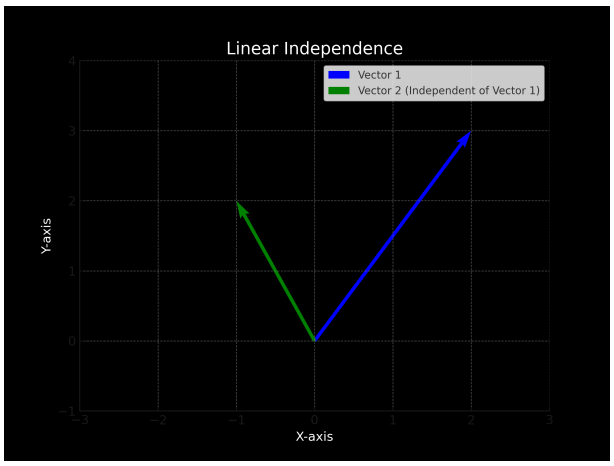


Figure 2.1: Linear Independence: Vectors $\mathbf{a}$ and $\mathbf{b}$. This graph shows two vectors (in blue and green) pointing in different directions. Their paths don't align, symbolizing their independence. Each vector stands on its own, not a scalar multiplication or addition of the other, embodying the concept of linear independence

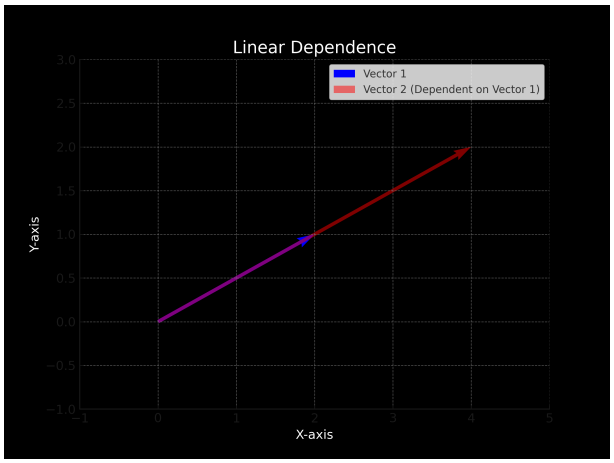


Figure 2.2: Linear Dependence: Vectors $\mathbf{v}$ and $\mathbf{w}$. This graph shows two vectors (in blue and green) pointing in different directions. Their paths don't align, symbolizing their independence. Each vector stands on its own, not a scalar multiplication or addition of the other, embodying the concept of linear independence

Chapter 3

Systems of Linear Equations

3.1 Introduction

A system of linear equations consists of two or more linear equations made up of two or more variables, such that all equations in the system are considered simultaneously. The solution to a system of linear equations is the set of values that satisfy all equations in the system simultaneously.

3.2 Definitions

3.2.1 Linear Equation

A linear equation in the variables $x_1, x_2, \ldots, x_n$ is an equation that can be written in the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$, where b and the coefficients $a_1, a_2, \ldots, a_n$ are real or complex numbers.

3.2.2 System of Linear Equations

A system of linear equations, or a linear system, is a collection of one or more linear equations involving the same set of variables. For example, a system of three equations in three variables x, y, and z might look like:

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1, \\ a_2x + b_2y + c_2z &= d_2, \\ a_3x + b_3y + c_3z &= d_3. \end{aligned}$$

3.3 Properties

In the context of linear algebra, the existence of a matrix inverse A^{-1} is closely tied to the nature of solutions to the system of equations $Ax = b$. For A^{-1} to exist, the system must have exactly one solution for every possible vector b. This condition indicates that A is a non-singular matrix, characterized by a non-zero determinant ($\det(A) \neq 0$).

3.3.1 Existence of Solutions

A system of linear equations may have:

- A unique solution,
- Infinitely many solutions, or
- No solution.

A system of equations is said to be consistent if it has at least one solution; otherwise, it is inconsistent.

3.3.2 Linear Combination of Solutions

If we have two solutions, x and y, to the system $Ax = b$, any linear combination of these solutions is also a solution. Mathematically, for any real number α, the vector $z = \alpha x + (1 - \alpha)y$ satisfies the system as well. This property is illustrated by the following:

$$A(\alpha x + (1 - \alpha)y) = \alpha Ax + (1 - \alpha)Ay = \alpha b + (1 - \alpha)b = b$$

The essence of this property lies in the principles of superposition inherent to linear systems. Because the system is linear, the principle of superposition applies: the weighted sum of two solutions is also a solution. This is due to two main properties of linear transformations:

1. Additivity: $A(x + y) = Ax + Ay$
2. Homogeneity: $A(\alpha x) = \alpha Ax$

Example

Consider a system of linear equations represented in matrix form as $Ax = b$, where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 11 \end{pmatrix}$$

Assume that $x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is a solution to $Ax = b$. Then

$$A \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 2 \\ 3 \cdot 1 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} = b$$

If $y = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is another solution, then

$$A \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 2 \cdot (-1) \\ 3 \cdot 2 + 4 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} = b$$

For a linear combination of x and y such that $z = \alpha x + (1 - \alpha)y$ with $\alpha = \frac{1}{2}$, we obtain

$$z = \frac{1}{2} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \left(1 - \frac{1}{2}\right) \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \end{pmatrix}$$

Verifying if z is a solution to $Ax = b$:

$$A \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 \cdot \frac{3}{2} + 2 \cdot \frac{1}{2} \\ 3 \cdot \frac{3}{2} + 4 \cdot \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{3}{2} + 1 \\ \frac{9}{2} + 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix} = b$$

Hence, z is indeed a solution to the system $Ax = b$. This principle holds true for any real α, showing the fundamental property of linear algebra that if two vectors are solutions to a system of linear equations with the same non-singular matrix A and vector b, then their linear combination is also a solution.

Significance and Intuition

The significance of this property is profound in linear algebra. It serves as the foundation for the concept of vector spaces, where vectors can be added together and multiplied by scalars to produce new vectors within the same space. This concept is also essential in understanding eigenvectors, eigenvalues, and diagonalization, which have vast applications in different fields such as physics, engineering, and computer science. Intuitively, you can think of each solution to the system $Ax = b$ as a specific "direction" or "path" that satisfies the equations. When we say that a linear combination of two solutions is also a solution, it's akin to saying that traveling along one "path" and then along another "path" still ends up at the destination defined by b. The ability to mix these "paths" freely without leaving the "landscape" defined by b is a powerful concept that reflects the core principles of linearity.
subsubsectionGraphical Interpretation of Solutions
The graphical representation provides an intuitive understanding of the solutions to the system of equations $Ax = b$. In the context of a two-dimensional space, solutions can be visualized as vectors originating from the origin. Each solution vector, when operated on by the matrix A, maps to the point b, which we refer to as the destination or result vector.
Consider the following graphical illustrations:

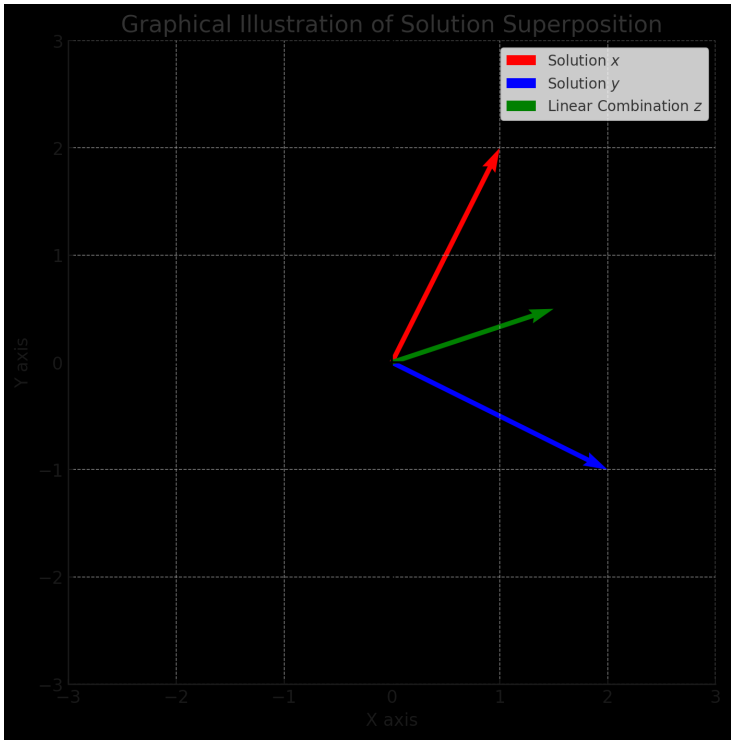


Figure 3.1: Vectors x and y representing two solutions to the system $Ax = b$, and z as their linear combination.

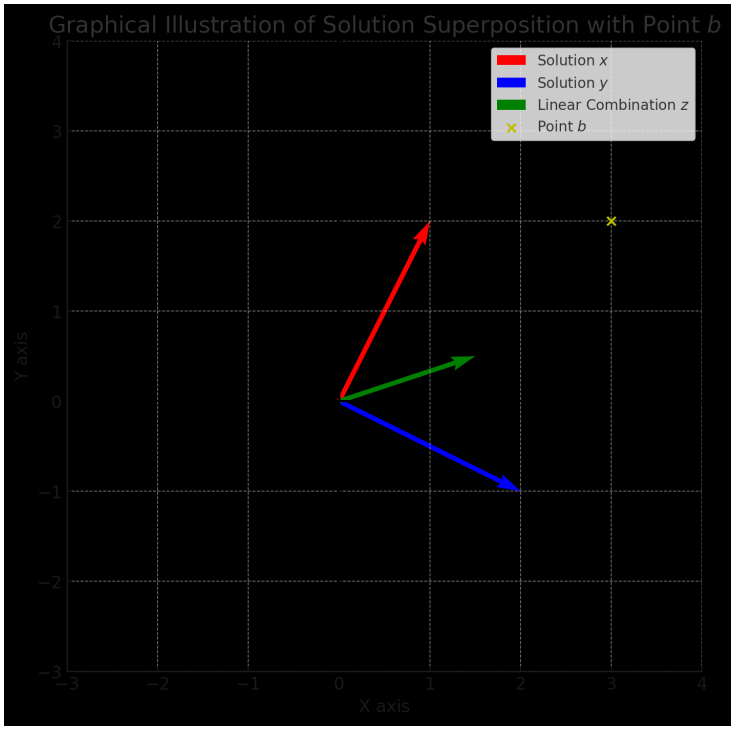


Figure 3.2: Point b represented in yellow, showcasing the destination that all solutions map to after being transformed by the matrix A.

To further elucidate, let us add a third illustration where the solution vectors x and y, as well as their linear combination z, are all connected by dotted lines to the destination point b. This emphasizes the path each vector takes to reach the destination after the transformation by A.

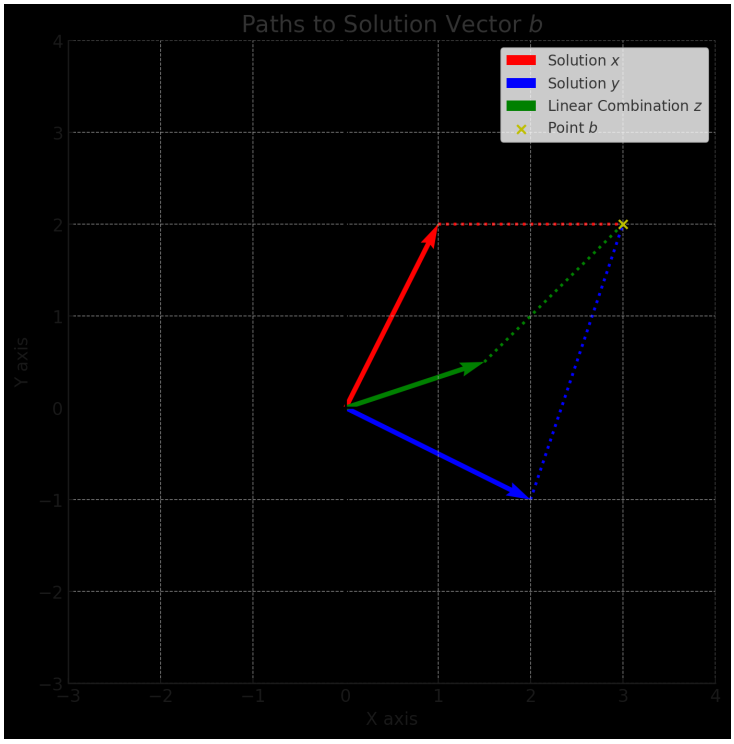


Figure 3.3: Dotted lines illustrating the paths taken by the solution vectors and their linear combination to reach the destination point b.

The dotted lines represent the conceptual paths that each solution would traverse in the vector space to arrive at b. The direction and length of each solution vector correspond to the specific "instructions" the matrix A applies to that vector to reach b. This reinforces the idea that a linear system can be viewed as a set of instructions for moving in a vector space, with the solution set forming a pathway that leads to the same endpoint, which is the vector b.
It is worth noting that the linearity of the system ensures that any scalar multiple or linear combination of the solutions will also follow a straight path to b, further illustrating the concept of superposition.

3.4 Examples with Solutions

3.4.1 Example 1: Unique Solution

Consider the system of equations:

$$x + y = 5, \tag{3.1}$$
$$2x - y = 1. \tag{3.2}$$

Solution:
Step 1: Add equations (3.1) and (3.2) to eliminate y.

$$\begin{aligned}(x + y) + (2x - y) &= 5 + 1\\ \Rightarrow 3x &= 6\\ \Rightarrow x &= 2.\end{aligned}$$

Step 2: Substitute $x = 2$ into equation (3.1) to solve for y.

$$\begin{aligned} 2 + y &= 5 \\ y &= 3. \end{aligned}$$

Thus, the solution is $x = 2$ and $y = 3$.

3.4.2 Example 2: No Solution

Consider the system:

$$x + y = 2, \tag{3.3}$$
$$x + y = 5. \tag{3.4}$$

Solution:
Step 1: Subtract equation (3.4) from equation (3.3).

$$\begin{aligned} (x + y) - (x + y) &= 2 - 5 \\ \Rightarrow 0 &= -3. \end{aligned}$$

The result is a contradiction, indicating that there is no possible value of x and y that can satisfy both equations simultaneously. Therefore, this system has no solution, demonstrating inconsistency.

3.4.3 Example 3: Using Matrix Multiplication

Consider the system of equations:

$$2x + 3y - z = 5, \tag{1}$$
$$x - 2y + 4z = 3, \tag{2}$$
$$-x + y + 2z = 2. \tag{3}$$

Solution:
Representing the system in matrix form $AX = B$, where

$$A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & -2 & 4 \\ -1 & 1 & 2 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix}.$$

Step 1: Calculate the inverse of matrix A, A^{-1}.
Step 2: Multiply A^{-1} by B to solve for X.

$$X = A^{-1}B.$$

Calculating A^{-1} and then multiplying by B provides the values of x, y, and z that satisfy the system of equations.

3.4.4 Example 4: Infinitely Many Solutions

Consider the system:

$$x + 2y - z = 0, \tag{4}$$
$$2x + 4y - 2z = 0. \tag{5}$$

Solution:
Representing the system in matrix form:

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The rows of A are linearly dependent, indicating that the system may have infinitely many solutions.

To find these solutions, we set one variable as a parameter (e.g., $y = t$). Then, we solve for x and z in terms of t:

$$x = -2t + k_1,$$
$$z = k_2,$$

where k_1 and k_2 are constants determined by the specific relationships between the equations in the system.

This process involves substituting the parameter back into the original equations and solving for the remaining variables, providing a set of solutions parameterized by t.

Observing A, we notice that the second row is a multiple of the first, indicating dependent equations and thus, potentially, infinitely many solutions.

To find solutions, we can set $y = t$ (where t is a parameter), and express x and z in terms of t.

These examples demonstrate the use of matrix operations to solve systems of linear equations, from finding unique solutions to identifying systems with infinitely many solutions or no solution at all.

Chapter 4

Eigendecomposition

4.1 Introduction to Eigendecomposition

Eigendecomposition is a mathematical process used extensively across various fields such as linear algebra, physics, computer science, and engineering. It involves decomposing a matrix into a set of its eigenvalues and eigenvectors, providing a powerful tool for understanding and solving linear transformations.

4.1.1 Intuition and Significance

At the heart of eigendecomposition is the concept of transforming vectors using matrices. When a matrix is multiplied by one of its eigenvectors, the output is simply the eigenvector scaled by a corresponding eigenvalue. This interaction unveils the essence of the matrix's transformation capabilities in multi-dimensional space.
The significance of eigendecomposition is widespread:

- Simplifies complex matrix operations and calculations.
- Serves as the foundation for Principal Component Analysis (PCA) in data science.
- Enables stability analysis in control systems through eigenvalues.
- Is fundamental in quantum mechanics for understanding quantum states.

4.1.2 Derivation of Eigenvalues and Eigenvectors

Given a square matrix A and a non-zero vector $\vec{x}$, $\vec{x}$ is an eigenvector of A if it is only scaled by a factor (the eigenvalue λ) when multiplied by A:

$$A\vec{x} = \lambda\vec{x}.$$

Rearranging gives us:

$$(A - \lambda I)\vec{x} = 0.$$

For non-trivial solutions, the determinant of $A - \lambda I$ must be zero:

$$\det(A - \lambda I) = 0.$$

Solving this characteristic equation provides the eigenvalues, λ, and substituting λ back into the equation allows for solving the eigenvectors, $\vec{x}$.

4.2 Examples

4.2.1 Finding Eigenvectors and Eigenvalues

Given matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Step 1: Determine the Characteristic Equation

The characteristic equation is found by setting the determinant of $A - \lambda I$ to zero:

$$\det(A - \lambda I) = \det\left(\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = 0$$

$$\det\left(\begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}\right) = 0$$

$$(4-\lambda)(3-\lambda) - (1)(2) = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

Step 2: Solve for Eigenvalues

Solving the quadratic equation $\lambda^2 - 7\lambda + 10 = 0$ gives:

$$\lambda_1 = 2, \quad \lambda_2 = 5$$

Step 3: Solve for Eigenvectors

For $\lambda_1 = 2$:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

One eigenvector is $\vec{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

For $\lambda_2 = 5$:

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Another eigenvector is $\vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

4.2.2 Eigendecomposition Example

Example 2:

With $\lambda_1 = 2$, $\lambda_2 = 5$, and eigenvectors $\vec{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$, $\vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, we demonstrate eigendecomposition.

Step 1: Formulate Q and Λ

$$Q = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

Step 2: Show $A = Q\Lambda Q^T$

$$Q\Lambda = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ -4 & 5 \end{pmatrix}$$

$$Q\Lambda Q^T = \begin{pmatrix} 2 & 5 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = A$$

1. With the eigenvalues and eigenvectors determined, matrix A can be represented as $A = VDV^{-1}$, where D is a diagonal matrix containing the eigenvalues, and V is a matrix whose columns are the corresponding eigenvectors.
2. To show $A = Q\Lambda Q^T$, we specifically arrange V (or Q in this notation) as the matrix of eigenvectors and Λ as the diagonal matrix of eigenvalues. This form emphasizes the orthogonality of eigenvectors in symmetric matrices, where $Q^{-1} = Q^T$.

In this detailed example, we explore how the eigendecomposition of matrix A is achieved, highlighting the process of breaking down a matrix into its fundamental components—eigenvalues and eigenvectors—through $A = Q\Lambda Q^T$.

This format provides a comprehensive overview of eigendecomposition, including its conceptual underpinnings, practical significance, and detailed examples that illustrate the process of finding eigenvalues, eigenvectors, and applying eigendecomposition to decompose a matrix.